Type MS · Interpret a raised or lowered blood cell count

Answer 1 of 4 · Question 5 part a.ii · 2 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

5 (a)	(i)	• 13 750 ÷ 7500; • 1:1.83;	Allow 1:1.8/1:2/ 1:1.8333... allow if they do it the other way round ie 7500 ÷ 13750	2
	(ii)	• person A has an infection/disease; • Immune response/more white blood cells needed, to produce antibodies/phagocytosis/destroy pathogen/virus/bacteria;	ignore sick ignore fight infection allow alternative correct explanation	2
	(iii)	Any two from: • excess bleeding; • poor/no blood clotting; • poor/slow wound healing/more time for pathogens to enter wound;		2
(b)	(i)	• tiredness/fatigue/high heart rate/faster breathing; • less oxygen carried/transported (to body cells); • less aerobic respiration; • less energy;		4
	(ii)	• HA is dominant; • parents are heterozygous; • HA expressed in heterozygous form; • so only needs one allele;	ignore ref to allele/gene for mpt 1	4

Total Marks 14

Type MS · Interpret a raised or lowered blood cell count

Answer 1 of 4 · Question 5 part a.ii · 2 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Interpret a raised or lowered blood cell count

Answer 2 of 4 · Question 1 part c.ii · 2 marks

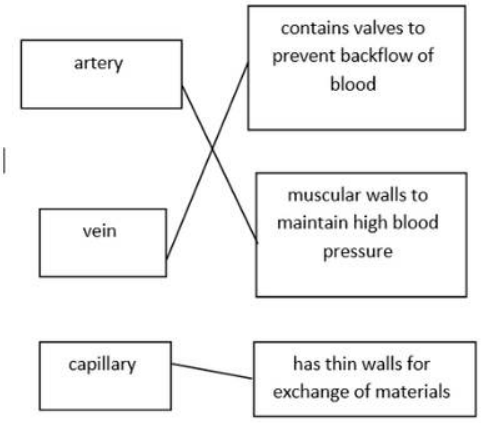
2020 NOV · Paper 01 (regional R) · find part c.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Interpret a raised or lowered blood cell count

Answer 3 of 4 · Question 2 part a.iii · 1 mark

2024 JUN · Paper 01 (regional R) · find part a.iii in this mark scheme extract

2 (a) (i)	X = plasma; Y = red blood cell; Z = white blood cell/phagocyte;	Accept named phagocyte	(3)
(a) (ii)	• transports oxygen; • to body/ cells/tissues/organs;		(2)
(a) (iii)	the patient has an infection/leukaemia/allergy/ blood cancer, inflammation;	Accept pathogens in body	(1)
(b)		Reject more than one line from each blood vessel 3 correct=2marks 1 or 2 correct= 1mark	(2)
(c) (i)	left ventricle;		(1)
(c) (ii)	• into right ventricle; • through pulmonary artery; • to the lungs; • (from the lungs) into the pulmonary vein;		(4)

Total for Question 2 = 13 marks

Type MS · Interpret a raised or lowered blood cell count

Answer 4 of 4 · Question 5 part a.iii · 2 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

5 (a)	(i)	• 13 750 ÷ 7500; • 1:1.83;	Allow 1:1.8/1:2/ 1:1.8333... allow if they do it the other way round ie 7500 ÷ 13750	2
	(ii)	• person A has an infection/disease; • Immune response/more white blood cells needed, to produce antibodies/phagocytosis/destroy pathogen/virus/bacteria;	ignore sick ignore fight infection allow alternative correct explanation	2
	(iii)	Any two from: • excess bleeding; • poor/no blood clotting; • poor/slow wound healing/more time for pathogens to enter wound;		2
(b)	(i)	• tiredness/fatigue/high heart rate/faster breathing; • less oxygen carried/transported (to body cells); • less aerobic respiration; • less energy;		4
	(ii)	• HA is dominant; • parents are heterozygous; • HA expressed in heterozygous form; • so only needs one allele;	ignore ref to allele/gene for mpt 1	4

Total Marks 14

Type MS · Interpret a raised or lowered blood cell count

Answer 4 of 4 · Question 5 part a.iii · 2 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks